

Appendix

1 NEUROUTE MODEL CONFIGURATIONS

Table 1 summarizes the encoder instantiations used in our evaluation. Across all settings, NeuRoute uses a lightweight MLP encoder that progressively compresses the input embedding from E_{dim} to a compact latent logit vector of dimension L_{dim} . Each hidden layer follows Linear→BatchNorm→ReLU, while the final encoder layer is linear to preserve the latent logits used for binarization and routing. We also report the dataset-specific τ_{emb} used for top-quantile pair selection; for a given dataset family (e.g., BigANN), we use the same τ_{emb} across scales.

2 TRAINING STABILITY AND CONVERGENCE

Figure 1 reports the training and validation loss curves for NeuRoute across all evaluated datasets under the default training protocol (2M training subset, batch size 4096, 500 epochs). Across datasets, the loss decreases rapidly in early epochs and then stabilizes, indicating consistent convergence behavior under a fixed schedule; we use this fixed schedule for all experiments for simplicity and reproducibility.

3 EMPIRICAL EVIDENCE FOR LOGIT-GUIDED BUCKET ROUTING

We focus on *bucket routing/refinement* rather than improving hash codes. Given fixed codes and a fixed hash-table layout, we exploit the pre-binarization logit margins to prioritize which buckets to probe under a limited budget. Figures 2, 3, and 4 support this design choice from three angles: (i) our bucket score correlates with KNN distance, (ii) useful candidates concentrate at small Hamming distances, and (iii) KNN bit flips are biased toward low-margin (uncertain) bits.

4 HASHING BASELINES AT 22 BITS: DIAGNOSTICS AND PURE-HASHING REFERENCE

We compare 22-bit codes produced by LSH, ITQ, TBH, and NeuRoute on BigANN (128D), Deep1B (96D), and GLDv2 (1536D). All methods use $b=22$ bits (3 bytes per vector), yielding $2^{22}=4,194,304$ buckets. We focus on two code-level diagnostics that affect any budgeted probing strategy: bucket balance (to control collision-driven candidate explosion) and margin-flip alignment under ground-truth neighbors (to justify margin-guided probing).

Table 3 reports training/encoding time and two bucket-quality metrics: *Nonempty%* (fraction of buckets with at least one item; higher indicates better utilization) and *MaxBucket%* (fraction of the database in the largest bucket; lower indicates fewer extreme hotspots). We report NeuRoute timing for completeness, but emphasize the structural bucket statistics that govern probe efficiency. NeuRoute is *not* treated as a pure hashing baseline in this subsection: we include its 22-bit codes only for diagnostics, while its end-to-end retrieval additionally uses budgeted traversal and centroid-stage filtering in the main paper.

Table 1: NeuRoute model design. Hidden layers use Linear → BatchNorm → ReLU; the final encoder layer is linear (no BatchNorm or activation).

Dataset	E_{dim}	L_{dim}	γ	τ_{emb}	Encoder widths
BigANN-100M	128	20	0.6	1.20	128, 96, 64, 20
BigANN-1B	128	22	0.6	1.20	128, 96, 64, 22
Deep1B-100M	96	20	0.6	0.85	96, 64, 32, 20
Deep1B-1B	96	22	0.6	0.85	96, 64, 32, 22
GLDv2	1536	16	0.6	0.90	1536, 512, 256, 128, 64, 16

Table 2: 22-bit hashing baselines (pure hashing, Hamming-only probing). CodeGen reports base code generation time. Best operating point is selected by maximum Recall@10 (ties broken by QPS).

Dataset	Method	CodeGen (s)	Codes (GiB)	Recall@10 (%)	QPS
BigANN-100M	LSH	26.8	0.279	7.84	1296
BigANN-100M	ITQ	90.1	0.279	11.04	2169
BigANN-100M	TBH	1006.1	0.279	26.12	239
Deep1B-100M	LSH	318.2	0.279	12.37	53702
Deep1B-100M	ITQ	77.6	0.279	35.41	3309
Deep1B-100M	TBH	997.6	0.279	81.10	41
GLDv2	LSH	391.5	0.026	37.55	83
GLDv2	ITQ	88.0	0.026	57.27	111
GLDv2	TBH	128.4	0.026	74.99	19

Table 3: Hash-code training/encoding cost and bucket statistics for 22-bit codes.

Method	Dataset	Train (s)	Encode (s)	Nonempty%	MaxBucket%
LSH	BigANN-100M	0.0	26.8	14.1	0.72
ITQ	BigANN-100M	15.0	90.1	52.4	0.56
TBH	BigANN-100M	1577.8	1006.1	0.64	1.18
NeuRoute	BigANN-100M	1176.5	63.0	59.8	0.02
LSH	BigANN-1B	0.0	375.3	23.8	0.72
ITQ	BigANN-1B	12.8	793.3	73.9	0.55
TBH	BigANN-1B	1577.8	9975.9	1.36	0.88
NeuRoute	BigANN-1B	1183.5	582.2	98.4	0.07
LSH	Deep1B-100M	0.0	318.2	81.5	0.14
ITQ	Deep1B-100M	14.5	77.6	66.8	0.87
TBH	Deep1B-100M	1538.1	997.6	0.018	8.83
NeuRoute	Deep1B-100M	1153.1	58.40	80.85	0.05
LSH	Deep1B-1B	0.0	4097.1	97.4	0.14
ITQ	Deep1B-1B	12.5	628.7	89.9	0.87
TBH	Deep1B-1B	1538.1	6122.8	0.024	8.83
NeuRoute	Deep1B-1B	1124.4	399.8	96.2	0.09
LSH	GLDv2	0.0	495.1	27.5	0.03
ITQ	GLDv2	59.2	88.4	16.1	1.34
TBH	GLDv2	2052.1	147.6	0.75	1.46
NeuRoute	GLDv2	2396.4	113.8	38.5	0.12

Bucket balance and collision structure. Figure 5 visualizes bucket occupancy distributions under 22-bit codes. Across datasets, better

Train/Val Loss

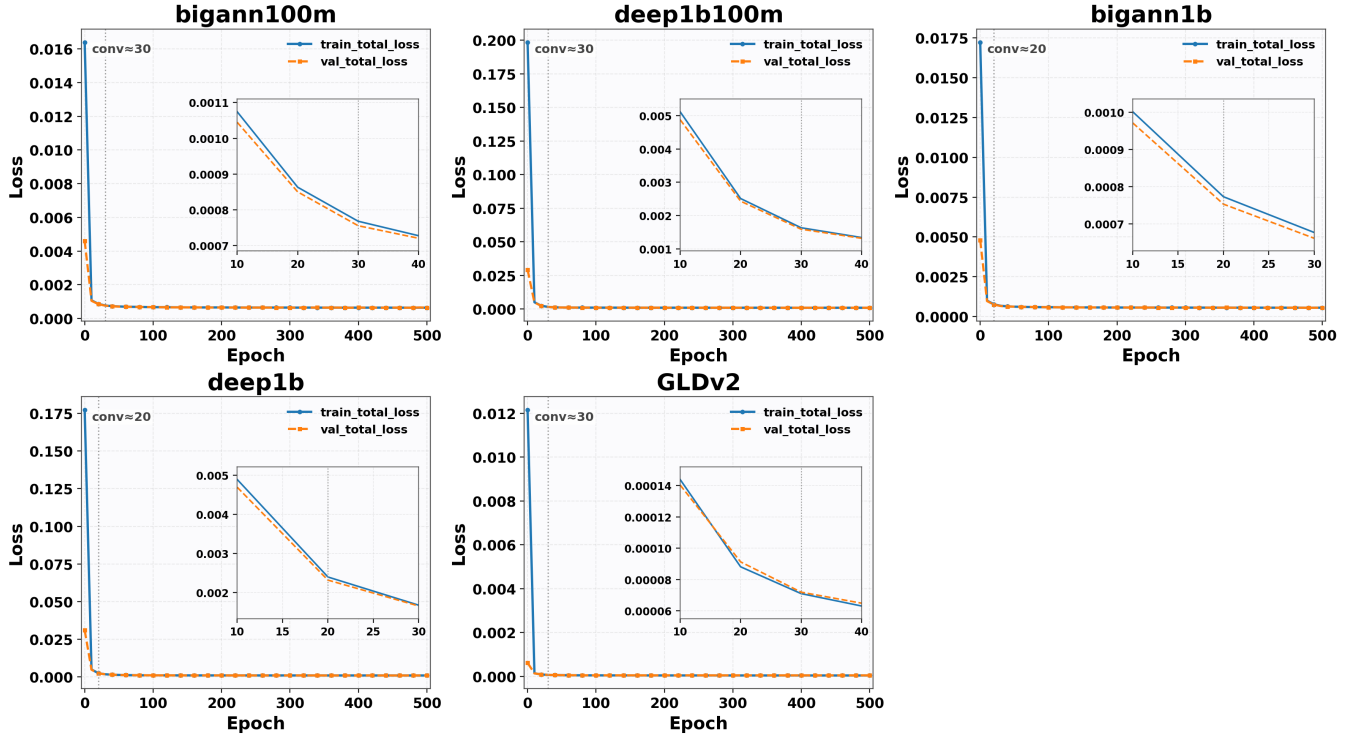


Figure 1: NeuRoute training/validation loss curves across datasets. Each panel shows the total loss over epochs with an inset zooming into the late-epoch regime to highlight convergence.

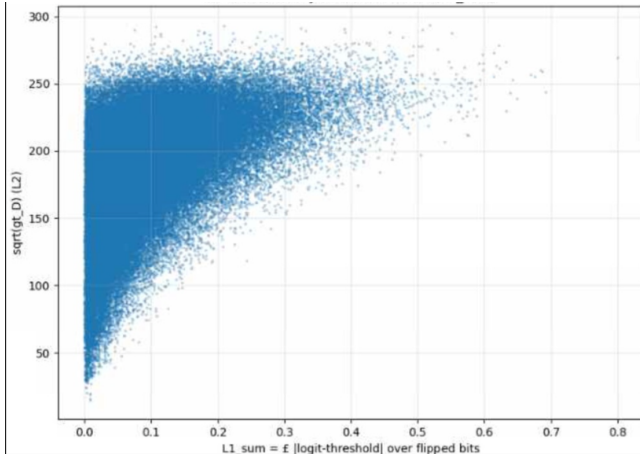


Figure 2: KNN embedding distance versus the proposed bucket score ($L1_sum$). Each point corresponds to a probed bucket; lower scores tend to be associated with closer GT neighbors.

bucket balance reduces the likelihood that a fixed probing budget is dominated by a small number of collision-heavy buckets.

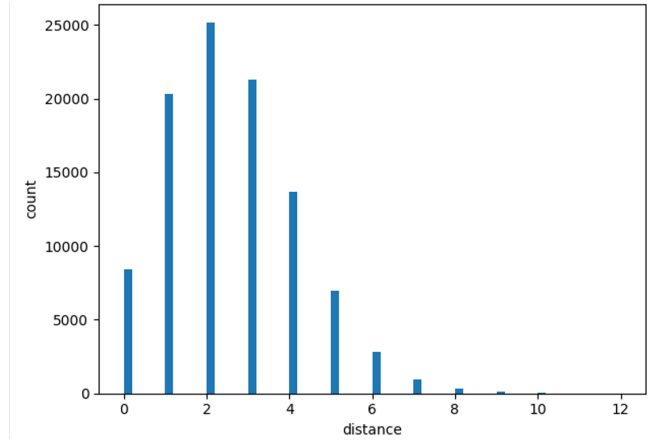


Figure 3: Distribution of query-candidate Hamming distances observed during probing. Useful candidates concentrate at small Hamming distances, motivating prioritized multi-bucket enumeration under a fixed budget.

Margin-flip alignment under ground-truth neighbors. Beyond bucket balance, margin-guided probing requires that ground-truth bit flips concentrate on low-margin (uncertain) bits. Figure 6 reports the flip-rank CDF diagnostic; across Deep1B and GLDv2, flips

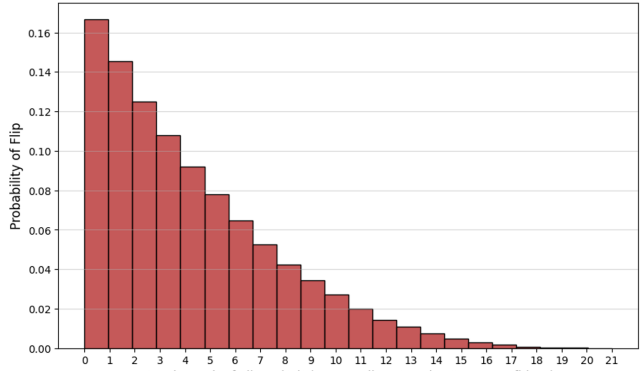


Figure 4: Probability that a KNN flips a bit as a function of the bit’s margin rank (rank 0 = smallest margin / least confident). GT flips concentrate on low-margin bits, validating margin-ranked bit flipping for bucket routing.

Method	Recall@10 (%)	QPS	Cand./q
LSH (22b)	37.55	82.94	72
ITQ (22b)	57.27	111.17	6,582
TBH (22b)	74.99	18.52	49,649
NeuRoute (ours)	93.86	111.40	69,709

Table 4: GLDv2 (1536D) comparison under the same evaluation harness ($Q=10,000$). Cand./q is derived from run logs (hashing baselines: `init_scanned_vecs/Q`; NeuRoute: `refined_vecs_phase2/Q`).

concentrate on low-margin bits, supporting the use of query-side margins as a principled signal for prioritized probing.

4.0.1 Pure hashing retrieval summary (22-bit). We summarize *pure hashing* baselines that rely on Hamming-only probing (LSH/ITQ/TBH). For space, we report the main comparison on GLDv2 (1536D) in Table 4.

5 CENTROID-STAGE BUDGET CONTROL

This section provides additional details for the centroid-stage gate and the heap-quality-driven early-stop mechanism used in Alg.1.

5.1 Margin-curve calibration for centroid-stage gating

Figure 7 visualizes the dataset-specific calibration of the margin curve $M(\cdot)$ used by the centroid-stage gate $\theta = d_{\min} + M(d_{\min})$. For each dataset, we bucket calibration samples by the current best centroid distance $d_{\min}$ (x-axis, “dbest2”) and estimate the empirical $q95/q99$ margin statistics at each knot (dashed curves). We then apply a smoothed, monotone “used” curve (solid) to make gating stable across knots; the bottom panel reports the number of samples per knot, indicating where the curve is well-supported.

Algorithm 1 α -driven early stopping in the centroid stage (single query).

Input: Query representation q ; binned bucket IDs `bins_buckets`; centroid index `subidx`; heap capacity K_c ; calibrated function $\alpha(\cdot)$; minimum bins $B_{\min}$; minimum no-gain batches $T_{\min}$; batch cap B_{cap} .

```

1: best  $\leftarrow +\infty$ 
2: consecutive_no_gain  $\leftarrow 0$ 
3: bins_processed  $\leftarrow 0$ 
4: Initialize an empty bounded heap  $\mathcal{H}$  with capacity  $K_c$ 
5: for each bin  $b$  in bins_buckets do
6:    $C \leftarrow \text{EXPANDBUCKETS\_TO\_CENTROIDS}(\text{subidx}, b.\text{bucket\_ids})$ 
7:   while  $C$  is not empty do
8:     batch  $\leftarrow \text{TAKE\_NEXT}(C, B_{\text{cap}})$ 
9:      $d^2 \leftarrow \text{DIST2}(q, \text{subidx}[\text{batch}])$ 
10:    best  $\leftarrow \min(\text{best}, \min(d^2))$ 
11:     $\alpha \leftarrow \alpha(\text{best})$   $\triangleright \alpha(\text{best}) \geq 0$ 
12:     $\theta \leftarrow \text{best} + \alpha$ 
13:    updates  $\leftarrow 0$ 
14:    for each  $(\text{cid}, \text{dist})$  in  $(\text{batch}, d^2)$  do
15:      if  $\text{dist} \leq \theta$  then
16:        if  $\text{HEAP\_PUSH}(\mathcal{H}, \text{dist}, \text{cid})$  succeeds then
17:          updates  $\leftarrow \text{updates} + 1$ 
18:        end if
19:      end if
20:    end for
21:    if updates = 0 then
22:      consecutive_no_gain  $\leftarrow$   $\leftarrow$ 
23:      consecutive_no_gain + 1
24:    else
25:      consecutive_no_gain  $\leftarrow 0$ 
26:    end if
27:    if  $\mathcal{H}$  is full and  $(\text{bins\_processed} + 1) \geq B_{\min}$  and
28:      consecutive_no_gain  $\geq T_{\min}$  then
29:      return  $\mathcal{H}$ 
30:    end if
31:  end while
32:  bins_processed  $\leftarrow \text{bins\_processed} + 1$ 
33: end for
34: return  $\mathcal{H}$ 

```

5.2 Heap-quality-driven early stopping in the centroid stage

Algorithm 1 details the α -driven early-stop criterion used in Alg.1. Given the current best centroid distance $d_{\min}$, we compute an acceptance band $\theta = d_{\min} + \alpha(d_{\min})$ and only admit centroids within this band into the bounded top- K_c heap. We stop early once the heap is full, a minimum number of bins have been processed, and a minimum number of consecutive batches yield no heap updates.

6 NEUROROUTE RETRIEVAL AND BUILD BREAKDOWN AT ~ 0.90 RECALL@10

This section reports NeuRoute’s query-time work decomposition at operating points with Recall@10 closest to 0.90 (Table 5), including

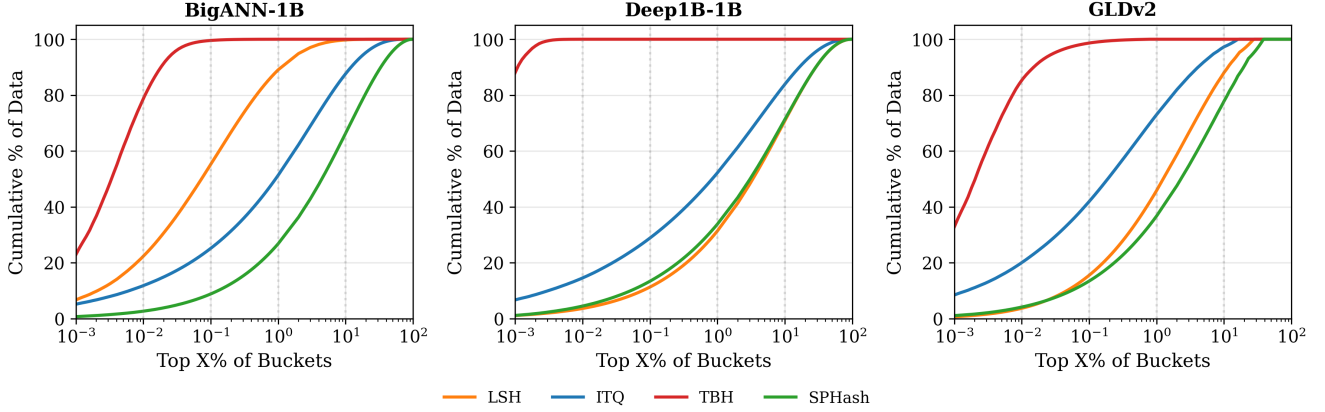


Figure 5: Bucket occupancy diagnostics for 22-bit codes on BigANN-1B, Deep1B-1B, and GLDv2 (LSH/ITQ/TBH/NeuRoute). Left: cumulative mass contained in the top $X\%$ buckets (sorted by occupancy). Right: top bucket mass by rank (top-1 to top-20).

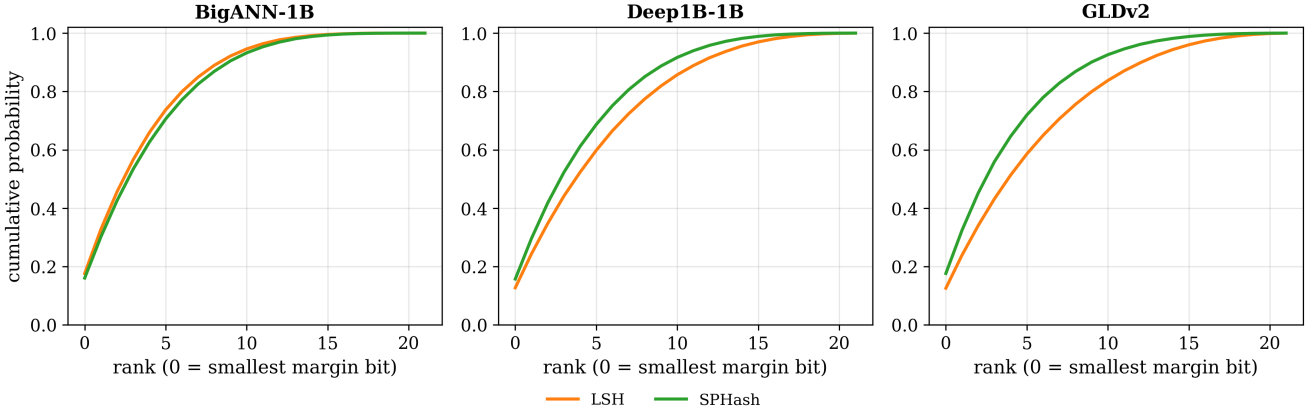


Figure 6: Flip-rank CDF (query-side) for 22-bit codes on BigANN-1B, Deep1B-1B, and GLDv2 (LSH vs. NeuRoute). For each query, bits are ranked by query-side margin (rank 0 is the smallest margin), and the CDF accumulates the ranks of bits that flip under XOR with ground-truth neighbors.

bucket-enumeration time, refinement time, and the resulting candidate volume per query. Together with the end-to-end build-time comparison in Fig. 8, these results highlight NeuRoute’s intended design point: budget-controlled retrieval with a lightweight time-to-index pipeline under a single-node setup.

7 BUCKET-LOCAL CLUSTERING WORKLOAD AND SCALING

7.1 Workload summary (1B-scale setting)

We report bucket-local clustering workload statistics for the 1B-scale setting in the encoded space ($N=1,000,000,000$ vectors; $L_{\text{dim}}=22$). The address space contains $n_{\text{list}} = 2^{L_{\text{dim}}} = 4,194,304$ buckets, of which 4,126,267 are non-empty (98.38%). We group buckets by size: large ($s \geq 129$): 1,313,975 buckets; small ($33 \leq s \leq 128$): 1,310,159 buckets; tiny ($s \leq 32$): 1,502,133 buckets. We apply k-means to large and small buckets (2,624,134 buckets); tiny buckets are not clustered (kept as single units). The resulting output scale is: total clusters

(capped) = 19,542,813; built sub-clusters $n_{\text{sub}} = 19,610,850$ (about 4.75 sub-clusters per non-empty bucket on average).

7.2 Per-bucket clustering policy

Let s denote bucket size and let $K_B(s)$ denote the number of clusters used for that bucket. We enforce a hard cap $K_B(s) \leq 2000$ for every bucket. For large buckets ($s \geq 129$), we set $K_B(s) = \min(2000, \lceil s/128 \rceil)$. For small buckets ($33 \leq s \leq 128$), we set $K_B(s) = \lceil s/32 \rceil$. For tiny buckets ($s \leq 32$), no k-means is performed; the bucket is kept as a single unit.

7.3 Clustering efficiency and thread scaling

We report clustering throughput as printed by the program (“approx = ... Mvec/s”) and measure scaling by varying the outer (bucket-parallel) thread count. Speedup and parallel efficiency are computed relative to the 8-thread run.

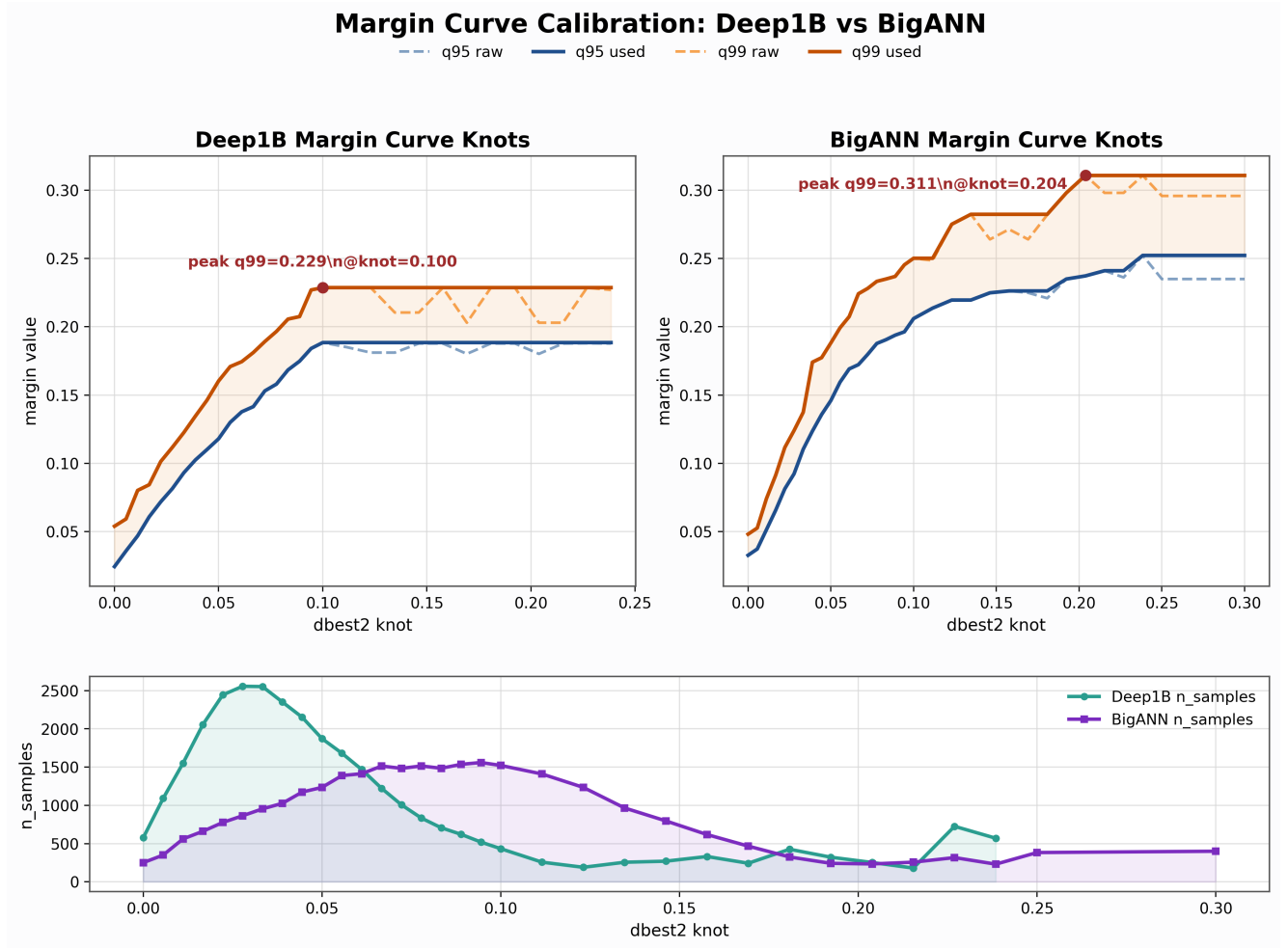


Figure 7: Margin curve calibration for centroid-stage gating on Deep1B-1B and BigANN-1B. Top: raw (dashed) and final used (solid) q_{95}/q_{99} margin knots as a function of $d_{\min}$ (“dbest2”). Bottom: number of calibration samples per knot for each dataset.

Table 5: NeuRoute retrieval breakdown at operating points with Recall@10 closest to 0.90 on each dataset. enum_ms, refine_ms, total_ms, total_qps, and recall are reported from query-level summary logs. model_ms = total_ms – enum_ms – refine_ms. Cand./q is the average number of refined candidates per query, computed as refined_vecs/Q.

Dataset	enum_ms	refine_ms	routing_ms	total_ms	total_qps	Cand./q	refined_vecs	recall
BigANN-1B	989.62	2832.63	31.51	3853.76	2594.87	193,591	1.94×10^9	0.89965
Deep1B-1B	1017.18	13022.58	45.46	14085.22	709.96	418,079	4.18×10^9	0.90000
BigANN-100M	268.98	1743.22	33.22	2045.42	4888.96	142,093	1.42×10^9	0.90059
Deep1B-100M	294.48	7110.12	38.04	7442.64	1343.61	227,449	2.27×10^9	0.90006
GLDV2	34.12	12718.24	220.81	12973.17	770.82	27,041	2.70×10^8	0.90066

Note. Sub-CSR construction after clustering is small (~ 5.5 – 6.4 s) relative to the clustering stage and does not affect the scaling trends.

8 GPU-ONLY REFERENCE (CAGRA)

We evaluated the GPU graph baseline CAGRA as a throughput reference. In our setup, CAGRA is limited by GPU memory and

can be constructed only up to 80M vectors on an A100 40GB, so we do not include it in the main Recall@10–QPS plots that focus on 100M/1B-scale comparisons.

At the 80M scale, CAGRA achieves very high throughput while maintaining strong recall: on BigANN-80M it reaches Recall@10 = 0.938 at $\sim 1.17 \times 10^6$ QPS, and on Deep-80M it reaches Recall@10 = 0.918 at $\sim 1.13 \times 10^6$ QPS. These results highlight the effectiveness

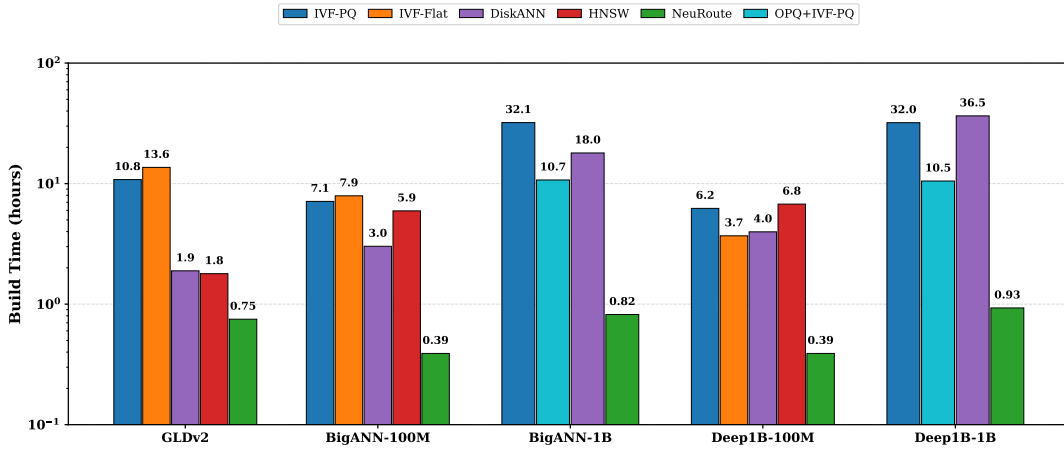


Figure 8: End-to-end index build time (hours, log-scale) for IVF-PQ, IVF-Flat, DiskANN, HNSW, OPQ + IVF-PQ, and NeuRoute.

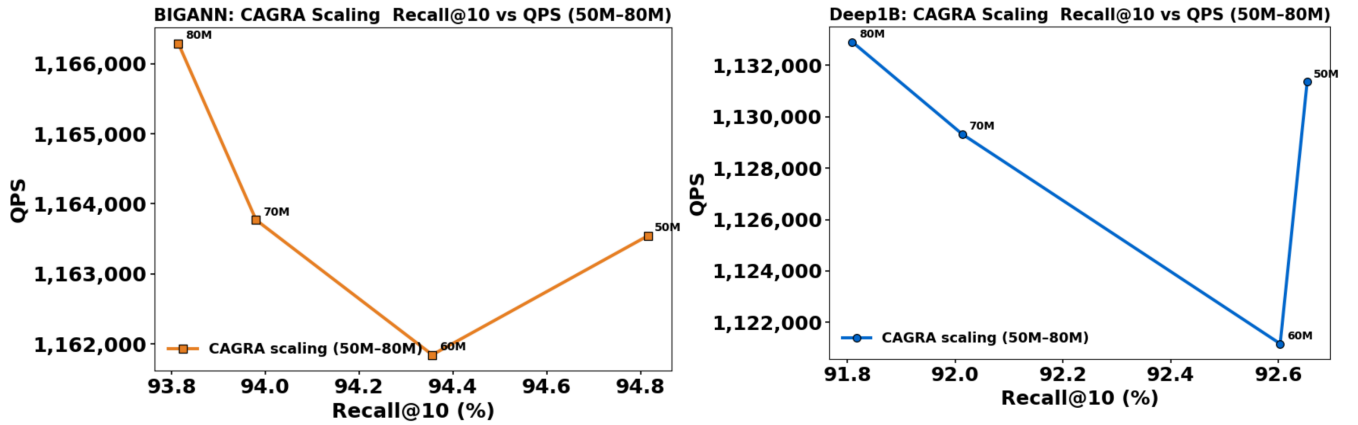


Figure 9: CAGRA GPU-only scaling reference at 50M-80M on BigANN and Deep1B. Points are labeled by dataset size (50M/60M/70M/80M).

Table 6: Clustering time and thread scaling for bucket-local clustering. Throughput is reported by the program as “approx = . . . Mvec/s”. Speedup and parallel efficiency are relative to the 8-thread run.

Threads	Time (s)	Throughput (Mvec/s)	Speedup vs. 8T	Efficiency vs. 8T
8	638.81	1.565	1.00×	100%
16	347.55	2.877	1.84×	91.9%
24	253.48	3.945	2.52×	84.0%

of GPU graph search under sufficient device memory, but its scaling is constrained in our environment.